# Beyond Sequence Metrics: State-Space Coverage and Algorithmic Design in Deterministic Channel Hopping

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**Abstract.** Frequency hopping is widely deployed in interference-prone wireless systems to mitigate fading and co-channel interference. Conventional sequence-level metrics, such as channel occupancy and Hamming correlation, evaluate emitted symbols but overlook complete controller contexts, including channel history and schedule phase, visited by a deterministic hopping policy. We model recurrent deterministic hopping through directed cycles in a finite controller-state graph, where the circumference $\mathrm{circ}(D)$ quantifies the maximum number of distinct contexts that can coexist in one recurrent orbit. We prove that an infinite family of strongly connected vertex-transitive digraphs satisfies $\mathrm{circ}(D) \leq (5/6)|V(D)|$, yielding a deterministic context-coverage gap $\gamma_{\mathrm{det}}(D) \geq 1/6$. This limitation persists despite full local symmetry and global reachability, revealing a global recurrence obstruction rather than a local capacity shortage. In graph-theoretic terms, the same construction yields a perimeter gap of at least $n/6$, which, to the best of our knowledge, is the strongest currently known linear bound for this class. For phase-dependent constraints, we introduce cyclic layering to expose layer-balance properties and matching bottlenecks, and develop mixed-integer formulations alongside a scalable cycle-cover-and-merge heuristic (LCCM). Experiments on phase-history models show that policies with nearly identical occupancy and Hamming profiles can exhibit context-coverage differences of up to 0.58, while LCCM achieves higher mean coverage than direct cycle-cover baselines with sub-second average runtimes. This framework separates structural recurrence limitations from resource bottlenecks and motivates multi-policy operation and selective constraint relaxation.

**Keywords.** channel hopping; context coverage; finite-state control; vertex-transitive digraphs; mixed-integer optimization.

**1 Introduction** Frequency hopping is widely used to mitigate interference and multipath fading in wireless communication systems. IEEE 802.15.4 time-slotted channel hopping (TSCH), which underlies 6TiSCH networks, combines communication in recurring slotframes with channel hopping [26, 25, 23]. Long-range frequency-hopping spread spectrum (LR-FHSS) employs fast frequency hopping to support long-range communication in large-scale IoT communication [3, 24].

The design of frequency-hopping sequences has been extensively studied [14, 19, 21]. Hamming autocorrelation and cross-correlation measure frequency hits between shifted sequences [14, 21]. No-hit-zone constructions impose restrictions on short-delay coincidences [29], while wide-gap constraints require a minimum separation between consecutive transmission frequencies [15, 30]. In rendezvous systems, the period is traded off against rendezvous diversity [6, 17]. These design criteria characterize properties of the transmitted channel sequence. They do not, in general, determine which complete internal states of the hopping controller are visited.

This distinction matters whenever hop selection depends on internal controller states beyond the current output symbol. In TSCH, for example, the physical channel is determined jointly by the absolute slot number and a channel offset [25, 26]. More generally, a finite-state hopping controller may record recent channel choices, an operating mode, or a scheduling phase. Identical channels can therefore arise across distinct controller contexts, so balanced channel usage does not guarantee traversal of all admissible states. We define context coverage as the proportion of admissible controller states visited in a single periodic orbit. In our evaluations, policy pairs with nearly indistinguishable occupancy imbalance and peak Hamming autocorrelation exhibit context-coverage disparities as large as 0.58.

Tracking channel histories connects closely to de Bruijn sequences and their constrained variants, which study the repetition of fixed-length symbol windows [7, 18]. When a controller state is restricted to a fixed-length channel history, context coverage reduces to the coverage of admissible histories. More broadly, our formulation incorporates operating modes and recurring phases as integral state dimensions.

This motivates the following question:

*Under the prescribed transition rules, what is*

*the maximum context coverage achievable in a single periodic orbit of a stationary deterministic policy?*

Let $D$ be the digraph whose vertices are complete controller states and whose arcs are admissible transitions. A stationary deterministic policy selects a fixed admissible successor for each state. Since the state space is finite, every trajectory eventually enters a directed cycle. Conversely, every directed cycle in $D$ can be realized as a periodic orbit of such a policy. Thus, the maximum number of distinct controller states visited in a single periodic orbit is the circumference $\mathrm{circ}(D)$, the length of a longest directed cycle. The maximum context coverage is therefore $\mathrm{circ}(D)/|V(D)|$. We define the *deterministic coverage gap* as $\gamma_{\mathrm{det}}(D) = 1 - \mathrm{circ}(D)/|V(D)|$. Formal definitions are given in Section 3.

The relation between symmetry and long cycles is a classical topic in graph theory. In the undirected setting, Babai established an $\Omega(\sqrt{n})$ lower bound on the circumference of connected vertex-transitive graphs on $n$ vertices [2]. For digraphs, Alspach asked whether the perimeter gap, defined as $n - \mathrm{circ}(D)$, can be arbitrarily large among connected vertex-transitive digraphs [1]. Bucić, Hendrey, Mohar, Steiner, and Yepremyan answered this question by constructing an infinite family with perimeter gap at least $(1 - o(1)) \ln n$, and conjectured that the gap can grow linearly with $n$ [5]. Li and Methuku subsequently confirmed this conjecture by constructing an infinite family with perimeter gap at least $n/12$ [16].

We improve this linear bound from $n/12$ to $n/6$. Specifically, we construct, for infinitely many $n$, a strongly connected vertex-transitive digraph $D$ on $n$ vertices with $\mathrm{circ}(D) \le 5n/6$, see Theorem 4.1. In the corresponding controller systems, every periodic orbit of a stationary deterministic policy therefore omits at least one sixth of the state space, giving $\gamma_{\mathrm{det}}(D) \ge 1/6$. Thus, strong connectivity and vertex-transitivity can coexist with a deterministic coverage gap bounded away from zero as the state space grows.

This limitation motivates methods for finding periodic orbits with large context coverage and bounding the maximum achievable coverage. For controllers with periodically varying transition constraints, we use a cyclically layered state-transition graph, a structure also used by Li and Methuku [16]. Each transition advances to the next phase, so every directed cycle visits the same number of states in each layer. Matchings between consecutive layers then yield upper bounds on coverage. We use these properties to develop exact mixed-integer linear programming (MILP) formulations and a layer-aware cycle-cover-and-merge (LCCM) heuristic. The heuristic adapts classical cycle patching [12] to the phase constraints of the controller graph.

Our contributions are summarized as follows.

1. **Sequence blind spots.** We show that conventional occupancy and Hamming criteria evaluate emitted channel symbols but can mask severe imbalances in the coverage of complete controller contexts. We therefore formulate deterministic hopping and related finite-state control problems as maximum directed-cycle coverage problems, using context coverage to expose this gap.
2. **Perimeter gap bound.** We construct an infinite family of strongly connected, vertex-transitive digraphs satisfying $\mathrm{circ}(D_M) \le 5M^2$ and $|V(D_M)| = 6M^2$, and hence having perimeter gap at least $M^2 = |V(D_M)|/6$.
3. **Layer-aware MILP.** We prove cyclic layer-balance equalities and matching upper bounds for phase-varying constraints, integrating them into exact mixed-integer formulations to diagnose and resolve structural bottlenecks.
4. **LCCM heuristic.** We design a scalable cycle-cover-and-merge heuristic based on first-return orbit structures, improving mean coverage from 0.657 to 0.761 on the $q = 4$ benchmark within sub-second runtimes.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the controller-state model and the coverage metrics. Section 4 presents the symmetric hopping family and the coverage-limit theorem. Section 5 develops the cyclic-layering theory, and Sections 6 and 7 give the exact formulations and the LCCM heuristic. Section 8 reports computational results, and Section 9 discusses scope and limitations.

## 2 Related Work

**2.1 Industrial Hopping** Channel hopping is a standard diversity mechanism in interference-prone wireless systems. TSCH uses time slots and channel hopping to mitigate multipath fading and co-channel interference [23, 26]. RFC 8180 specifies a minimal 6TiSCH configuration with a recurring slotframe and a channel hopping sequence [25]. LR-FHSS uses fast frequency hopping for long range, large-scale IoT communication [3, 24]. In all of these systems a controller executes hopping repeatedly, and the next hop can depend on recent channel decisions, a protocol mode, and a schedule phase. System evaluation, however, still focuses on the emitted channel sequence.

**2.2 Sequence Metrics and Constrained Sequence Design** Classical frequency hopping sequence design evaluates observable channel strings. For a peri-

odic sequence $c_0, \dots, c_{L-1}$, channel occupancy is

$$p_f = \frac{1}{L} \sum_{t=0}^{L-1} \mathbf{1}\{c_t = f\}, \tag{2.1}$$

and the normalized periodic Hamming autocorrelation at shift $d$ is

$$H(d) = \frac{1}{L} \sum_{t=0}^{L-1} \mathbf{1}\{c_t = c_{t+d \bmod L}\}. \tag{2.2}$$

Hamming autocorrelation and cross correlation quantify repeated hits under shifts [14, 21]. No-hit-zone and nonrepeating constructions suppress short delay coincidences [29]. Adjacent symbol and wide gap constraints enforce local frequency separation [15, 30]. Constrained de Bruijn constructions study cyclic objects under local window restrictions [7, 8, 18].

These methods treat the emitted word as the design object. The combinatorial goal is typically an Eulerian representation of all allowed words or edges. We instead optimize a single simple cycle in the controller state graph. A sequence can therefore attain good values of (2.1) and (2.2) while the generating cycle omits many admissible contexts. Section 8 exhibits policies with nearly identical occupancy and Hamming profiles whose context coverages differ by a factor of more than three.

**2.3 Long Cycles in Symmetric Graphs** Long cycles in vertex-transitive graphs have been studied for decades. In the undirected setting, the conjectures of Lovász and Thomassen predict, respectively, the existence of Hamiltonian paths in all connected vertex-transitive graphs and Hamiltonian cycles in all sufficiently large ones. Babai proved that every connected vertex-transitive graph on $n$ vertices contains a cycle of length greater than $\sqrt{3n}$ [2]. This long-standing bound was successively improved to $(1 - o(1))n^{3/5}$ by DeVos [9], to $\Omega(n^{13/21})$ by Groenland et al. [11], and to $\Omega(n^{9/14})$ by Norin et al. [20]. Most recently, Bucić, Christoph, Pokrovskiy, and Steiner obtained a cycle of length $n^{2/3-o(1)}$ [4]. These results concern undirected graphs and do not directly extend to directed cycles.

The directed problem has a different history, beginning with work of Rankin on Cayley digraphs [22]. Vertex-transitive digraphs need not be Hamiltonian [27], and Alspach asked whether their perimeter gap, namely the difference between the order and the circumference, can be arbitrarily large [1]. Bucić, Hendrey, Mohar, Steiner, and Yepremyan recently answered this question by constructing infinitely many vertex-transitive digraphs with perimeter gap at least $(1 - o(1)) \log n$; they also proved that every connected vertex-transitive digraph on $n$ vertices contains a directed cycle of length $\Omega(n^{1/3})$ [5]. Li and Methuku subsequently strengthened both directions: they constructed infinitely many such digraphs with perimeter gap at least $n/12$ and proved the general lower bound $\Omega(\sqrt{n})$ for the circumference [16].

Classical Hamiltonicity criteria for digraphs operate in a substantially denser regime. For example, the theorem of Ghouila-Houri guarantees a Hamilton cycle under minimum semidegree at least $n/2$ [10], while Woodall's Ore-type condition requires $d^+(u) + d^-(v) \geq n$ whenever $uv$ is not an arc [28]. For sufficiently large oriented graphs, Keevash, Kühn, and Osthus proved the sharp condition $\delta^+(D), \delta^-(D) \geq (3n-4)/8$ [13]. These results do not apply to the sparse regime considered here: the Cayley digraphs in Section 4 have outdegree two (and indegree two), and they are not oriented because some arcs occur in antiparallel pairs.

Our construction therefore lies in the sparse, highly symmetric regime. It gives an infinite family of strongly connected vertex-transitive Cayley digraphs $D$ of outdegree two satisfying $\mathrm{circ}(D) \leq 5|V(D)|/6$. Equivalently, their perimeter gap is at least $|V(D)|/6$, improving the previously known explicit linear gap $|V(D)|/12$ of Li and Methuku by a factor of two.

**2.4 Cycle Covers and Patching** Relaxing a single cycle requirement to a cycle cover and then merging components is a classical idea in traveling salesman heuristics [12]. Our graphs have additional structure to which generic patching does not directly apply. The time expanded graph consists of cyclic phase layers $X_0 \to X_1 \to \cdots \to X_{T-1} \to X_0$, so every feasible recurrent orbit selects the same number of states from each layer (Proposition 5.1). A balanced cover can therefore be built layer by layer through bipartite matchings, and merging is governed by the orbit structure of the first return permutation. The LCCM heuristic in Section 7 is organized around this fact.

**3 Controller-State Model** Let $\mathcal{F}$ be the finite set of available channels. We model a hopping controller as a finite deterministic transition system

$$\mathcal{T} = (S, \mathcal{A}, \Omega, f, \mathcal{F}, \chi),$$

where $S$ is the finite state space, $\mathcal{A}$ is the action set, $\Omega \subseteq S \times \mathcal{A}$ is the set of admissible state–action pairs, $f : \Omega \to S$ is the transition function, and $\chi : S \to \mathcal{F}$ is the output map. For each $s \in S$, let $\mathcal{A}(s) := \{a \in \mathcal{A} : (s, a) \in \Omega\}$, and assume that $\mathcal{A}(s) \neq \varnothing$ for every $s \in S$.

The underlying transition digraph of $\mathcal{T}$ is $D_{\mathcal{T}} = (S, E)$, where $E := \{(s, f(s, a)) : (s, a) \in \Omega\}$. We assume that the state representation is Markovian: the next state is determined by the current state and action,

and the output is determined by the current state.

A stationary deterministic policy is a map $\pi : S \to \mathcal{A}$ such that $\pi(s) \in \mathcal{A}(s)$ for every $s \in S$. It induces the deterministic transition map $f_\pi : S \to S$ defined by $f_\pi(s) := f(s, \pi(s))$. A trajectory under $\pi$ is a sequence $(s_t)_{t\geq 0}$ satisfying $s_{t+1} = f_\pi(s_t)$. The functional digraph of $f_\pi$ has outdegree one at every vertex. Since $S$ is finite, each weakly connected component contains a unique directed cycle, and every trajectory eventually enters that cycle. Conversely, every directed cycle in $D_\mathcal{T}$ is a periodic orbit under some stationary deterministic policy.

DEFINITION 3.1 (Deterministic single-orbit state coverage). *Let $D$ be a finite transition digraph. Its circumference is*

$$\mathrm{circ}(D) := \max\big\{|V(C)| : C \text{ is a directed cycle in } D\big\}.$$

*The deterministic state-coverage ratio and the corresponding state-coverage gap of $D$ are defined by* $\rho_{\det}(D) := \dfrac{\mathrm{circ}(D)}{|V(D)|}$ *and* $\gamma_{\det}(D) := 1 - \rho_{\det}(D)$.

Thus $\gamma_{\det}(D) = 0$ if and only if $D$ contains a directed Hamilton cycle. Equivalently, some stationary deterministic policy induces a periodic orbit visiting every controller state. If $\gamma_{\det}(D) > 0$, then every periodic orbit visits at most $(1-\gamma_{\det}(D))|V(D)|$ states and hence omits at least $\gamma_{\det}(D)|V(D)|$ states. Thus $\gamma_{\det}(D)$ measures a structural limitation of the transition graph that is distinct from physical-layer metrics such as throughput or packet delivery ratio. Figure 3.1 illustrates the shift from sequence-level observation to state-space coverage.

To make the abstraction concrete, consider a controller with two-symbol channel memory, $q := |\mathcal{F}| \geq 3$ available channels, and phase period $m$. A state is

$$s_t = (c_{t-1}, c_t, \tau_t),$$

where $c_{t-1}, c_t \in \mathcal{F}$ satisfy $c_{t-1} \neq c_t$ and $\tau_t \in \mathbb{Z}_m$. A transition

$$(c_{t-1}, c_t, \tau) \longrightarrow (c_t, c_{t+1}, \tau + 1 \bmod m) \tag{3.1}$$

is legal whenever $c_{t+1} \notin \{c_{t-1}, c_t\}$. The output map is $\chi(c_{t-1}, c_t, \tau) = c_t$. Hence the transition graph has $mq(q-1)$ vertices and outdegree $q-2$.

The case $q = 4$ is the smallest one in which every state has more than one admissible successor. It has $12m$ vertices and outdegree two, coincides with the cyclically layered construction of Li and Methuku [16], and serves as a canonical test case in our computational study.

**4 Structural Coverage Limit** Section 3 introduced deterministic state coverage in a general controller-state graph and illustrated it with a concrete phase-history model. That model is motivated by hopping constraints, but its graph structure depends on the particular choices of channel count, memory depth, and phase period. We now separate these application-specific features from the structural question of interest: can a positive deterministic coverage gap persist even when the state graph is strongly connected and vertex-transitive?

To answer this question, we introduce a family of Cayley digraphs. The construction is chosen to enforce strong connectivity and vertex-transitivity while isolating a global obstruction to a state-spanning directed cycle. The resulting family provides the structural coverage bound established below.

**4.1 Symmetric Hopping Family** Let $r \geq 2$, let $M = 2^r$, and define

$$A = \mathbb{Z}[\omega]/M\mathbb{Z}[\omega], \qquad \omega^2 + \omega + 1 = 0. \tag{4.1}$$

As an additive group, $A \cong \mathbb{Z}_M^2$, so every element of $A$ can be written uniquely as $u + v\omega$ with $u, v \in \mathbb{Z}_M$. Depending on the application, the two coordinates may index, for example, a frequency group and a code/offset class, or a subband and a beam class.

Set $\lambda = 1 + \omega$. Since $\lambda^2 = \omega$, $\lambda^3 = -1$ and $\lambda^6 = 1$, multiplication by $\lambda$ defines an automorphism of $A$ of order 6. Let $\mathbb{Z}_6$ act on $A$ through this automorphism, and define

$$\Gamma = A \rtimes \mathbb{Z}_6. \tag{4.2}$$

Thus the group operation is

$$(x, i)(y, j) = (x + \lambda^i y, i + j), \tag{4.3}$$

where the second coordinate is taken modulo 6.

The controller state is

$$s = (x, i) \in \Gamma, \tag{4.4}$$

where $x$ is the logical resource index and $i$ is a six-valued mode index. Hence the state space has cardinality $|\Gamma| = 6M^2$. Define two generating actions by $a = (0, 2)$ and $b = (1, 3)$. Right multiplication by these elements gives

$$T_a(x, i) = (x, i)a = (x, i + 2) \tag{4.5}$$

and

$$T_b(x, i) = (x, i)b = (x + \lambda^i, i + 3). \tag{4.6}$$

Action $a$ changes the mode index while leaving the logical resource index fixed, whereas action $b$ changes

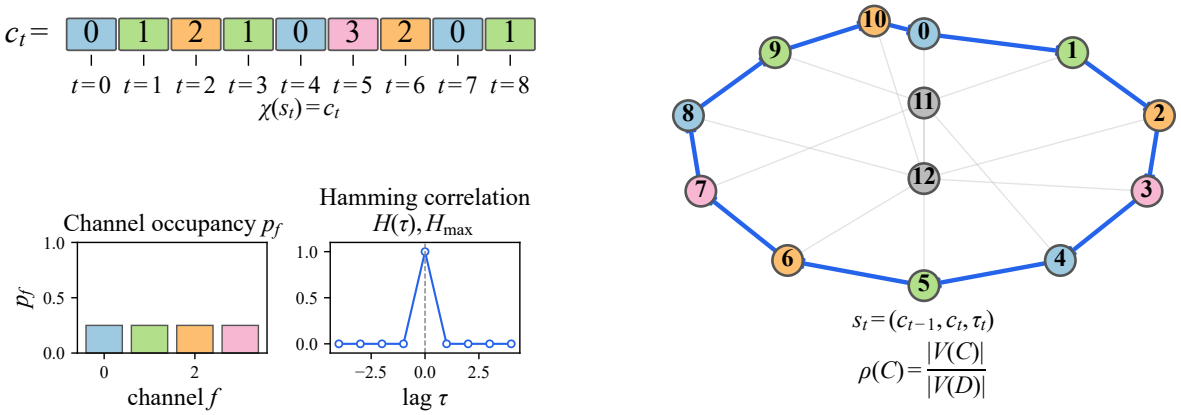



Figure 3.1: Observable sequence metrics characterize emitted channel symbols, whereas the controller-state representation captures complete history–mode–phase contexts. The output map is many-to-one, so different controller contexts may produce the same observable channel. Under a stationary deterministic policy, each recurrent orbit is a directed cycle in the state graph.

both the resource index and the mode index. The controller state graph is the directed Cayley graph

$$D_M = \mathrm{Cay}(\Gamma; \{a, b\}), \tag{4.7}$$

with vertex set $\Gamma$ and arcs $g \to ga$ and $g \to gb$ for every $g \in \Gamma$.

An application-specific output map $\chi : \Gamma \to \mathcal{F}$ assigns an observable channel or resource label to each controller state, and a trajectory $(s_t)$ produces the output sequence $c_t = \chi(s_t)$. The structural result below depends solely on the state graph $D_M$ and is independent of the particular choice of $\chi$. A concrete output map is therefore required only when channel-level quantities such as occupancy, Hamming correlation, or packet delivery performance are evaluated.

### 4.2 Coverage Bound

THEOREM 4.1 (Symmetric deterministic state-coverage limit). *For every $r \geq 2$ with $M = 2^r$, the state graph $D_M$ defined by (4.5)–(4.6) is strongly connected and vertex-transitive, has $6M^2$ vertices, and satisfies*

$$\mathrm{circ}(D_M) \leq 5M^2. \tag{4.8}$$

*Thus,*

$$\rho_{\det}(D_M) \leq \frac{5}{6}, \qquad \gamma_{\det}(D_M) \geq \frac{1}{6}. \tag{4.9}$$

*Proof sketch.* Direct calculations give $a^3 = b^2 = (0,0)$, $b(aba^{-1}) = (1-\omega, 0)$, and $a\big(b(aba^{-1})\big)a^{-1} = (\omega(1-\omega), 0)$. The last two elements generate all translations in $A$: they generate $(1-\omega)\mathbb{Z}[\omega]$ additively, and multiplication by $1-\omega$ is invertible modulo $M$ because $(1-\omega)(1-\omega^2) = 3$ and $M$ is a power of 2. Since the images of $a$ and $b$ generate $\mathbb{Z}_6$, we have $\langle a, b\rangle = \Gamma$. Moreover, $a^{-1} = a^2$ and $b^{-1} = b$, so $D_M$ is strongly connected; left multiplication by $\Gamma$ proves vertex-transitivity.

The $a$-arcs form directed triangles. Contracting them turns the $b$-arcs into the bipartite graph $H$ with vertices $E_x, O_x$ $(x \in A)$ and edges $E_xO_{x+1}$, $E_xO_{x+\omega}$, and $E_xO_{x+\omega^2}$. This is a $2M^2$-vertex toroidal quotient of the hexagonal tiling. A directed cycle using both arc types meets each contracted triangle at most once, because every visit uses at least two of its three vertices, and hence projects to a simple cycle $Q$ in $H$. Write $\ell = |Q|$, and let $u$ count the visits using two $a$-arcs rather than one. Then $|C| = 2\ell + u$, while the total signed turn of $Q$ is $(2u - \ell)\pi/3$.

If $Q$ is non-contractible, its turning number is zero, so $u = \ell/2$ and $|C| = (5/2)\ell \leq 5M^2$. If $Q$ is contractible, its planar lift has total turn $\pm 2\pi$, so $|2u - \ell| = 6$. Such a $Q$ cannot be Hamiltonian: otherwise its boundary would contain all $2M^2$ vertices of $H$, and Euler's formula together with the hexagonal face count would give $f = (M^2 - 1)/2$, impossible because $M$ is even. Since $H$ is bipartite, $\ell \leq 2M^2 - 2$, and therefore $|C| \leq (5\ell + 6)/2 \leq 5M^2 - 2$. The cycles using only $a$-arcs or only $b$-arcs have lengths 3 and 2. Full details of the toroidal embedding and turn calculation appear in Appendix A.1. ∎

Table A.1 reports computational checks on the first three finite instances. For $r = 1$ and $r = 2$, the MILP certifies $L = 5M^2$. For $r = 3$, the solver reaches its time

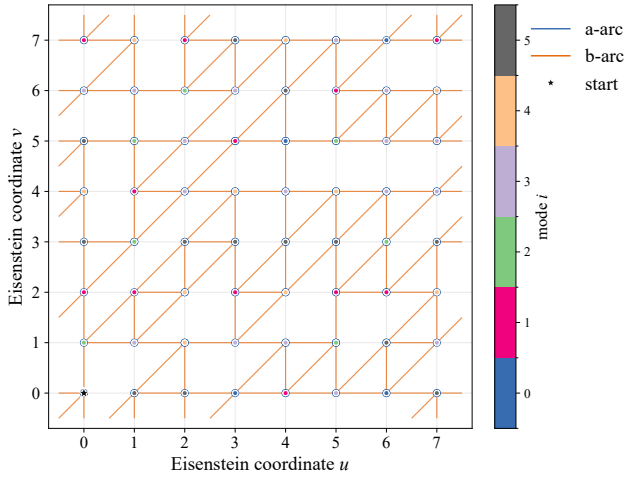



Figure 4.1: Cayley cycle trajectory for $r = 3$ with $M = 8$ and 384 vertices, projected onto the Eisenstein coordinate plane. The cycle visits 320 states and omits 64, with omitted states appearing across all six mode layers.

limit, while an independently verified cycle has length $320 = 5M^2$.

Theorem 4.1 shows that strong connectivity and vertex transitivity do not guarantee complete deterministic recurrent coverage. The gap cannot be explained by unreachable states or by a locally defective region. Rather, it arises from a global obstruction to forming a spanning directed cycle. The Cayley family provides the analytic construction for this structural result, while the phase history model of Section 3 serves as the computational model for protocol constraints. The bound in Theorem 4.1 applies strictly to the Cayley family. Figure 4.1 illustrates this global obstruction. In the displayed instance, the cycle visits 320 of the 384 states, with omitted states appearing across all six mode layers.

Theorem 4.1 establishes an inherent limitation of stationary deterministic recurrence on this graph. The next subsection preserves the underlying state graph while varying solely the policy class, demonstrating the contrasting coverage dynamics that emerge under randomized action selection.

**4.3 Randomized Exploration** Theorem 4.1 relies on the fixed successor choice of a stationary deterministic policy. To see what changes when this restriction is removed, we keep the same state graph and consider independent randomized action selection. This gives the Markov chain $(X_t)_{t\geq 0}$ on $\Gamma$ with transition probabilities

$$P(g, ga) = P(g, gb) = \frac{1}{2}, \quad \text{for every } g \in \Gamma. \tag{4.10}$$

Proposition 4.2 (Uniform randomized exploration). *For every $r \geq 2$ with $M = 2^r$, the Markov chain defined by (4.10) is irreducible and aperiodic. Its unique stationary distribution is the uniform distribution*

$$\mu(g) = \frac{1}{6M^2}, \qquad g \in \Gamma. \tag{4.11}$$

*Moreover, for every initial state $g \in \Gamma$,*

$$\lim_{t\to\infty} \left\| P^t(g, \cdot) - \mu \right\|_{\mathrm{TV}} = 0, \tag{4.12}$$

*and every controller state is visited infinitely often with probability one.*

*Proof.* The strong connectivity of $D_M$ established in Theorem 4.1 implies irreducibility. Since $a^3 = (0, 0)$ and $b^2 = (0, 0)$, every state lies on closed walks of lengths 3 and 2. The period of the chain therefore divides $\gcd(2, 3) = 1$, so the chain is aperiodic.

Right multiplication by either $a$ or $b$ is a permutation of $\Gamma$. Hence the transition matrix $P$ is the average of two permutation matrices and is therefore doubly stochastic. The uniform distribution is consequently stationary. Since the state space is finite and the chain is irreducible, this stationary distribution is unique and every state is recurrent. Aperiodicity gives convergence to the stationary distribution, while recurrence implies that every state is visited infinitely often with probability one. ∎

Together, Theorem 4.1 and Proposition 4.2 show that the coverage limitation is specific to deterministic recurrence. On the same state graph, a stationary deterministic policy omits at least one sixth of the states, whereas randomized action selection visits every state infinitely often with probability one. This distinction concerns state-space exploration and does not by itself imply improved physical-layer performance.

**5 Cyclic Layering** We now return to deterministic recurrent policies. The phase-history model of Section 3 and time-expanded versions of the Cayley family of Section 4 share a common cyclic structure. For a recurring period $T$, the state set is partitioned into layers $X_0, \ldots, X_{T-1}$. For each $\tau \in \mathbb{Z}_T$, let $E_\tau$ denote the set of admissible transitions from $X_\tau$ to $X_{\tau+1 \bmod T}$. Thus

$$E_\tau \subseteq X_\tau \times X_{\tau+1 \bmod T}.$$

Hence

$$X_0 \to X_1 \to \cdots \to X_{T-1} \to X_0. \tag{5.1}$$

We write $D^{[T]}$ for such a graph.

For the phase-history model, $T = m$. A time-expanded Cayley graph has vertex set $V(D_M) \times \mathbb{Z}_T$ and transitions

$$(g, \tau) \to (gu, \tau + 1 \bmod T), \qquad u \in \{a, b\}.$$

Phase-dependent constraints remove arcs from the sets $E_\tau$ without changing this layered structure. The results below depend on this structure.

### 5.1 Layer and Matching Bounds

PROPOSITION 5.1 (Layer balance). *Every directed cycle $C$ in a cyclically layered graph $D^{[T]}$ has length $L = hT$ for some integer $h$, and*

$$|C \cap X_0| = \cdots = |C \cap X_{T-1}| = h. \tag{5.2}$$

*Proof.* Every selected arc increases the phase by one modulo $T$, so a return to the starting phase needs $L \equiv 0 \pmod T$; during any $T$ consecutive steps a trajectory visits each phase once, which gives equal occupancy. □

A longest-cycle solver can therefore impose phase balance explicitly rather than rediscover it through branch-and-bound. For each boundary $X_\tau \to X_{\tau+1}$, let $B_\tau$ be the bipartite graph of legal lifted transitions and $\nu(B_\tau)$ its maximum matching size.

PROPOSITION 5.2 (Matching bottleneck). *For every cyclically layered instance,*

$$\operatorname{circ}(D^{[T]}) \le T \min_\tau \nu(B_\tau). \tag{5.3}$$

*Proof.* A cycle with $h$ states per layer induces a one-to-one set of $h$ transitions across every boundary, so $h \le \nu(B_\tau)$ for all $\tau$; with (5.2), $|C| = hT$. □

Equation (5.3) is an upper bound, not a Hamiltonicity certificate. Writing the layer deficiency

$$\delta_\tau = |X_\tau| - \nu(B_\tau), \tag{5.4}$$

a positive $\delta_\tau$ is a *local capacity failure*: too few compatible transitions cross that boundary. By contrast, $\delta_\tau = 0$ for every $\tau$ does not guarantee a spanning recurrent orbit, since local perfect matchings may compose into several disjoint cycles.

**5.2 First-Return Structure** Fix an integer $h$ with $1 \le h \le \min_\tau |X_\tau|$. A balanced matching system of order $h$ consists of subsets $Y_\tau \subseteq X_\tau$ with $|Y_\tau| = h$, together with bijections $M_\tau : Y_\tau \to Y_{\tau+1}$ (indices modulo $T$) whose arcs $(y, M_\tau(y))$ are admissible in $D^{[T]}$. Its first-return permutation is

$$\Pi := M_{T-1} \circ \cdots \circ M_1 \circ M_0 : Y_0 \longrightarrow Y_0. \tag{5.5}$$

PROPOSITION 5.3 (First-return decomposition). *If the cycle decomposition of $\Pi$ has cycle lengths $r_1, \ldots, r_k$, then the selected matching arcs form a directed cycle factor on $\bigcup_{\tau \in \mathbb{Z}_T} Y_\tau$ consisting of exactly $k$ directed cycles, of lengths $r_1 T, \ldots, r_k T$. In particular, the selected arcs form a single directed cycle of length $hT$ if and only if $\Pi$ is a single $h$-cycle.*

*Proof.* Each $y \in Y_\tau$ has exactly one selected outgoing arc (via $M_\tau$) and one selected incoming arc (via $M_{\tau-1}$), so the arcs form a directed cycle factor. Following the arcs from $y \in Y_0$ through all $T$ layers reaches $\Pi(y) \in Y_0$, and a return to $X_0$ occurs only after a multiple of $T$ steps; hence an orbit of length $r$ under $\Pi$ lifts to a directed cycle of length $rT$, and distinct orbits yield vertex-disjoint cycles. □

Consequently, a balanced cover forms a single recurrent orbit if and only if its first-return permutation is a single cycle. Hence perfect matching capacity at every boundary ($\delta_\tau = 0$) is necessary but not sufficient for state-complete recurrence. The orbit count $\kappa(\Pi)$ measures the fragmentation of the selected cycle factor. When $\kappa(\Pi) > 1$, the selected matchings produce multiple periodic orbits, but another matching choice may still yield a single orbit.

**5.3 Structural Obstructions** Assume that $|X_\tau| = n$ for every $\tau$, so that $|V(D^{[T]})| = nT$. The preceding results give a natural hierarchy of obstructions to a state-spanning recurrent orbit. The first two provide explicit graph-level certificates and may occur simultaneously. If both tests pass, any remaining failure is a global compatibility obstruction among the boundary matchings. For a boundary with deficiency $\delta^*$, Hall's theorem gives a set $U \subseteq X_\tau$ such that

Table 5.1: Structural obstructions to a state-spanning recurrent orbit.

| Obstruction | Condition / diagnostic | Coverage implication |
|---|---|---|
| Local matching deficiency | $\delta^* = \max_\tau \delta_\tau > 0$; Hall-deficient set | $\gamma_{\det} \ge \delta^*/n$ |
| Loss of strong connectivity | Largest SCC order $s_{\max} < nT$ | $\gamma_{\det} \ge 1 - s_{\max}/(nT)$ |
| Residual first-return obstruction | $\delta^* = 0$, $s_{\max} = nT$, but no system of perfect boundary matchings makes $\Pi$ an $n$-cycle | $\gamma_{\det} \ge 1/n$ |

$$|U| - |N_{B_\tau}(U)| = \delta^*.$$

Proposition 5.2 then gives the first coverage bound. The second bound follows because every directed cycle lies inside a strongly connected component.

If both tests pass, Proposition 5.3 gives the remaining condition. A state-spanning orbit exists if and only if some choice of perfect boundary matchings makes $\Pi$ a single $n$-cycle. If $\kappa(\Pi) > 1$ for one matching system, that system contains several cycles. This does not rule out another matching system with $\kappa(\Pi) = 1$. If no such system exists, every recurrent orbit uses at most $n-1$ states from each layer. Therefore

$$\gamma_{\det} \geq \frac{1}{n}.$$

**6 Optimization** For a state graph $D = (V, E)$ with nonnegative context weights $w_v$, define

(6.1)
$$L^\star(D) = \max\{|C| : C \text{ is a directed simple cycle}\},$$

(6.2)
$$U^\star(D) = \max_C \sum_{v \in C} w_v.$$

The first objective maximizes state coverage. The second gives different values to different states through the weights $w_v$. These weights can represent utility, importance, or testing priority. The formulation uses a fixed root $r_0$. When a canonical root can be used, one root is sufficient. Otherwise, different roots are handled outside the model. Constraint (6.3g) is used only in the layer aware variant.

(6.3a)
$$\max \sum_{v \in V} y_v \quad \left(\text{or } \sum_{v \in V} w_v y_v\right)$$

(6.3b)
$$\text{s.t.} \sum_{u:(u,v)\in E} x_{uv} = y_v, \quad \sum_{w:(v,w)\in E} x_{vw} = y_v, \qquad \forall v \in V,$$

(6.3c)
$$y_{r_0} = 1,$$

(6.3d)
$$\sum_{u:(u,v)\in E} f_{uv} - \sum_{w:(v,w)\in E} f_{vw} = y_v, \qquad \forall v \neq r_0,$$

(6.3e)
$$\sum_{w:(r_0,w)\in E} f_{r_0 w} - \sum_{u:(u,r_0)\in E} f_{u r_0} = \sum_{v \neq r_0} y_v,$$

(6.3f)
$$0 \leq f_{uv} \leq (|V|-1)x_{uv}, \quad \forall (u,v) \in E,$$

(6.3g)
$$\sum_{v \in X_\tau} y_v = \sum_{v \in X_0} y_v, \quad \tau = 1, \ldots, T-1,$$

(6.3h)
$$x_{uv}, y_v \in \{0,1\}, \; f_{uv} \geq 0.$$

Table 6.1: MILP notation.

| | |
|---|---|
| $V, E$ | states and admissible transitions |
| $X_\tau$ | phase layer $\tau$ |
| $w_v$ | context weight of state $v$ |
| $x_{uv}$ | selected-arc indicator |
| $y_v$ | selected-state indicator |
| $f_{uv}$ | connectivity flow on arc $(u, v)$ |
| $r_0$ | fixed connectivity root |

Degree constraints (6.3b) ensure that the selected arcs form a directed cycle cover. The root (6.3c) and flow constraints (6.3d)–(6.3f) route one unit of flow from $r_0$ to each selected non-root state. This eliminates disconnected subtours, ensuring that all selected states form a single strongly connected component without requiring full state selection. In the layer-aware variant, constraint (6.3g) communicates the equal phase occupancy of Proposition 5.1 directly to the solver. Its effect on solver performance is evaluated in Section 8. Finally, the objective (6.3a) maximizes either unweighted coverage $\sum_v y_v$ or weighted utility $\sum_v w_v y_v$.

**7 LCCM Heuristic** The exact formulation in Section 6 searches directly for a single cycle, whereas LCCM scales to larger instances by exploiting cyclic layering. The method operates across three successive stages comprising cover construction, orbit merging, and repair or reinsertion. For a target occupancy $h$, LCCM matches consecutive layers and retains complete first-return orbits, falling back to direct subset sampling if this fails. By Proposition 5.3, cycles in the cover correspond to orbits of the first-return permutation $\Pi$, making the reduction of $\kappa(\Pi)$ the primary objective.

To merge components, LCCM greedily applies admissible two-edge exchanges at single phase boundaries to decrease $\kappa(\Pi)$ within a search budget. When merging stalls with $\kappa(\Pi) > 1$, repair deletes a complete orbit and rematches remaining states at reduced occupancy. Conversely, reinsertion attempts occupancy $h+1$ whenever capacity permits, accepting the candidate only if its largest component strictly expands. Algorithm 7.1 outlines the complete procedure.

For a balanced cover $\mathcal{M}$, let $h(\mathcal{M})$ denote its occupancy per layer and let $\kappa(\mathcal{M})$ denote the number of cycles in the first-return permutation induced by $\mathcal{M}$.

**Algorithm 7.1** Layer-aware cycle-cover-and-merge (LCCM)

**Require:** Layered graph $D$; target occupancy $\bar{h}$ (default: smallest layer size); move limit $b$; stage options and budgets; seed
**Ensure:** A directed cycle $C$ or construction failure

Build a balanced cover $\mathcal{M}$ at occupancy $\bar{h}$
**if** construction fails **then**
  **return** construction failure
**end if**
$\mathcal{M} \leftarrow \mathrm{Merge}(\mathcal{M}, b)$
$b' \leftarrow \max\{1, \lfloor b/2 \rfloor\}$
**if** repair enabled, $\kappa(\mathcal{M}) > 1$, $h(\mathcal{M}) > 1$, and time remains **then**
  Delete the chosen whole orbit and rematch the remaining states, obtaining a candidate $\mathcal{M}'$
  **if** a candidate exists and time remains **then**
    $\mathcal{M}' \leftarrow \mathrm{Merge}(\mathcal{M}', b')$
    **if** the candidate satisfies the repair acceptance rule **then**
      $\mathcal{M} \leftarrow \mathcal{M}'$
    **end if**
  **end if**
**end if**
**if** reinsertion enabled, $h(\mathcal{M}) < \min_\tau |X_\tau|$, and time remains **then**
  Build a candidate $\mathcal{M}'$ at occupancy $h(\mathcal{M}) + 1$
  **if** a candidate exists **then**
    $\mathcal{M}' \leftarrow \mathrm{Merge}(\mathcal{M}', b')$
    **if** time remains and the candidate satisfies the reinsertion acceptance rule **then**
      $\mathcal{M} \leftarrow \mathcal{M}'$
    **end if**
  **end if**
**end if**
**return** the largest component of the terminal cover $\mathcal{M}$

LCCM provides no approximation guarantee. A construction failure does not certify that a longer directed cycle is absent. Implementation details and search budgets are given in Appendix A.4.

**8 Computational Results** We evaluate the phase-history model of Section 3 with $q = |\mathcal{F}|$ channels, phase period $m$ (corresponding to $T$ in the layered formulation), and memory depth $k = 2$. An unrestricted instance has $mq(q-1)$ states and outdegree $q-2$. For each returned cycle $C$, we report the coverage ratio $\rho = |V(C)|/|V(D)|$, a feasible lower bound on $\rho_{\det}(D)$. The evaluation spans $q \in \{4, 6, 8\}$, $m \in \{16, 32, 64\}$, twenty seeds per setting, and seven constructors. Tables report unit-weight means with two-sided 95% confidence intervals over seeds. Full protocol details, additional cohorts, and timing provenance appear in Appendix A.3.

**8.1 Phase-History Benchmark** We begin with the smallest nontrivial phase history model introduced in Section 3. It has a two-symbol channel memory with $k = 2$ and $q = 4$ available channels. The phase period is $m$. This graph has $12m$ states and outdegree two. This specialization coincides with the cyclically layered construction of Li and Methuku [16]. We use it as the canonical benchmark for the exact formulation and LCCM.

On this model, the cut-free canonical MILP returns a cycle of length $L = 11m$ for thirteen periods from $m = 2$ to 64. The solver itself certifies optimality for $m = 2, 3, 4$. For the remaining periods, the MILP terminates without closing the solver gap; however, optimality of the $11m$ incumbent follows from the Li–Methuku upper bound [16], since this phase-history graph coincides with their $12m$-vertex layered construction and every directed cycle has length at most $11m$. Thus $L = 11m$ is the exact circumference for all listed periods. Figure 8.1 shows an independent $m = 8$ replay with an 88-state cycle.

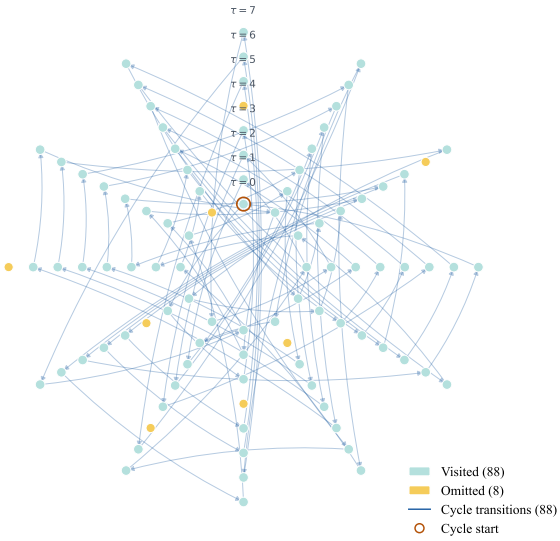



Figure 8.1: Validated $m = 8$ phase-history trajectory. The cycle visits 88 states and omits 8 states across eight phase layers, matching the $11m$ incumbent length.

**8.2 Coverage and Generalization** Table 8.1 compares LCCM-full with the largest cycle in an initial cover, greedy extension, and random admissible extension. LCCM-full has the highest mean coverage at every tested setting. At $q = 4$, its mean coverage ranges from

0.746 to 0.775. At $q = 6$ and $q = 8$, every LCCM-full run reaches full coverage and exceeds the strongest baseline mean by 0.193 to 0.330. These results show that the partial coverage observed at $q = 4$ is not a general limitation of the phase history family. One greedy extension run at $(q, m) = (4, 64)$ returns no valid cycle and is excluded from the coverage statistics.

Figure 8.2 shows how these results vary with the phase period. LCCM-full remains above all baselines at $q = 4$ and reaches full coverage at $q = 6$ and $q = 8$. Greedy extension and random admissible extension generally lose coverage as $m$ increases, whereas LCCM-full does not show this decline.

The ablation in Appendix A.6 explains why. At $q = 4$, no two edge exchange is accepted during merging, so the improvement after initialization comes entirely from orbit guided repair and reinsertion. Removing orbit guidance lowers mean coverage by 0.088 to 0.133, and the paired 95% intervals exclude zero at all three periods. At $q = 6$ and $q = 8$, all four variants reach full coverage, so the saturated results do not isolate a separate contribution from each component. Additional ablation tables, initializer comparisons, and search traces appear in Appendices A.6 through A.8.

**8.3 Sequence Metrics** We now ask whether conventional sequence summaries detect the coverage differences observed above. Using the $q = 4$ subset with $m \in \{16, 32\}$, we compute an occupancy-imbalance statistic $\Delta_{\mathrm{occ}} = \max_f |p_f - 1/q|$ and a Hamming autocorrelation peak $H_{\max} = \max_{1 \le d < L} H(d)$ from (2.1)–(2.2). We compare all pairs sharing $(q, m, \text{weight mode})$ that satisfy

$$\begin{aligned} |\Delta_{\mathrm{occ}}^{(i)} - \Delta_{\mathrm{occ}}^{(j)}| &< 0.02, \\ |H_{\max}^{(i)} - H_{\max}^{(j)}| &< 0.03, \\ |\rho_i - \rho_j| &> 0.10. \end{aligned} \tag{8.1}$$

Of 9660 eligible unordered pairs, 2092 satisfy this rule. The largest coverage difference is $7/12 \approx 0.583$, between cycles of coverage 0.8333 and 0.2500 whose occupancy-imbalance difference is 0.01875 and Hamming-peak difference is about 0.00208. Figure 8.3 shows how either summary can conceal this difference. The two policies are similar under both sequence summaries yet differ threefold in context coverage. Featured pairs are listed in Appendix A.9.

**8.4 Scalability and Structural Diagnostics** Table 8.2 compares MILP and LCCM across seven phase periods at $q = 4$. Exposing layer balance reduces the $m = 4$ solve time from 16.1 to 3.2 seconds (Appendix A.5). For $m \ge 8$ the solver times out with a 9.1% gap, while LCCM-full returns cycles in sub-second mean time. The best heuristic runs match the recorded MILP incumbent $L/|V| = 11/12$ at every listed period.

For $m = 4, 8, 16$, the deterministic cycles cover $11/12$ of the states. Uniform random walks on the same instances eventually exceed this coverage. The mean numbers of steps needed are 112.8, 213.9, and 448.7. Further results are given in Appendix A.10.

We also test 200 perturbed $q = 4$ instances with transition deletion and channel blacklists. Among them, 125 have positive matching deficiency and 122 are not strongly connected. Eight valid returned covers contain more than one first return orbit. Some instances have more than one of these problems. Valid coverage results are obtained for 75 instances. The other 125 runs fail to construct a cover within the search budget. Matching deficiency and loss of strong connectivity provide direct checks for the failure modes in Table 5.1.

**9 Discussion** Standard metrics such as channel occupancy and Hamming correlation focus strictly on emitted symbols. Our Cayley construction proves a state-space recurrence gap, while the experiments show that occupancy and Hamming summaries may fail to detect such gaps. Even when a graph is strongly connected and vertex-transitive, every deterministic periodic policy must miss at least one sixth of the states. Full symmetry and reachability do not guarantee full state exploration. Our experiments show that two policies can have nearly identical occupancy and Hamming values while their context coverage values differ by as much as 0.583.

The computational results show how transition rules shape coverage. Larger state spaces are not necessarily harder to cover. In the four-channel case, each state has an outdegree of two. This low degree makes the graph rigid, so simple edge swaps cannot merge disconnected cycles. Orbit-guided repair and reinsertion are therefore necessary to gain coverage. In contrast, instances with six or eight channels have outdegrees of four or six. Higher connectivity gives enough freedom for simple two-edge exchanges to merge cycle covers into full cycles. These findings indicate that coverage gaps come from low node degrees and tight memory rules rather than graph size.

Cyclic layering gives engineers a practical diagnostic direction. When coverage is low, one can inspect layer balance and boundary matchings to identify the structural case. A positive matching deficiency $\delta_\tau > 0$ means that too few transitions cross a phase boundary. System designers can fix this local shortage by relaxing constraints or allowing more channels. If every boundary has a perfect matching but the first-return map remains broken into several orbits, the issue is structural. In that case, instead of searching for a single cycle, the

Table 8.1: Primary heuristic results. Coverage is mean ± 95% confidence interval over 20 unit-weight seeds, except greedy extension at $(q, m) = (4, 64)$, which has 19 valid cycles. Time is mean LCCM-full runtime.

| $q$ | $m$ | $\lvert V\rvert$ | $d^+$ | LCCM-full | Cover cycle | Greedy | Random | LCCM time (s) |
|---|---|---|---|---|---|---|---|---|
| 4 | 16 | 192 | 2 | $0.762 \pm 0.077$ | $0.629 \pm 0.103$ | $0.579 \pm 0.027$ | $0.504 \pm 0.039$ | 0.032 |
| 4 | 32 | 384 | 2 | $0.775 \pm 0.061$ | $0.688 \pm 0.095$ | $0.504 \pm 0.030$ | $0.417 \pm 0.036$ | 0.121 |
| 4 | 64 | 768 | 2 | $0.746 \pm 0.064$ | $0.654 \pm 0.091$ | $0.478 \pm 0.018$ | $0.333 \pm 0.022$ | 0.508 |
| 6 | 16 | 480 | 4 | $1.000 \pm 0.000$ | $0.608 \pm 0.087$ | $0.748 \pm 0.009$ | $0.732 \pm 0.027$ | 0.109 |
| 6 | 32 | 960 | 4 | $1.000 \pm 0.000$ | $0.580 \pm 0.099$ | $0.707 \pm 0.010$ | $0.622 \pm 0.018$ | 0.367 |
| 6 | 64 | 1920 | 4 | $1.000 \pm 0.000$ | $0.615 \pm 0.088$ | $0.670 \pm 0.010$ | $0.560 \pm 0.026$ | 1.672 |
| 8 | 16 | 896 | 6 | $1.000 \pm 0.000$ | $0.569 \pm 0.082$ | $0.807 \pm 0.008$ | $0.804 \pm 0.015$ | 0.338 |
| 8 | 32 | 1792 | 6 | $1.000 \pm 0.000$ | $0.588 \pm 0.083$ | $0.779 \pm 0.009$ | $0.716 \pm 0.016$ | 1.407 |
| 8 | 64 | 3584 | 6 | $1.000 \pm 0.000$ | $0.646 \pm 0.089$ | $0.762 \pm 0.008$ | $0.655 \pm 0.015$ | 5.398 |

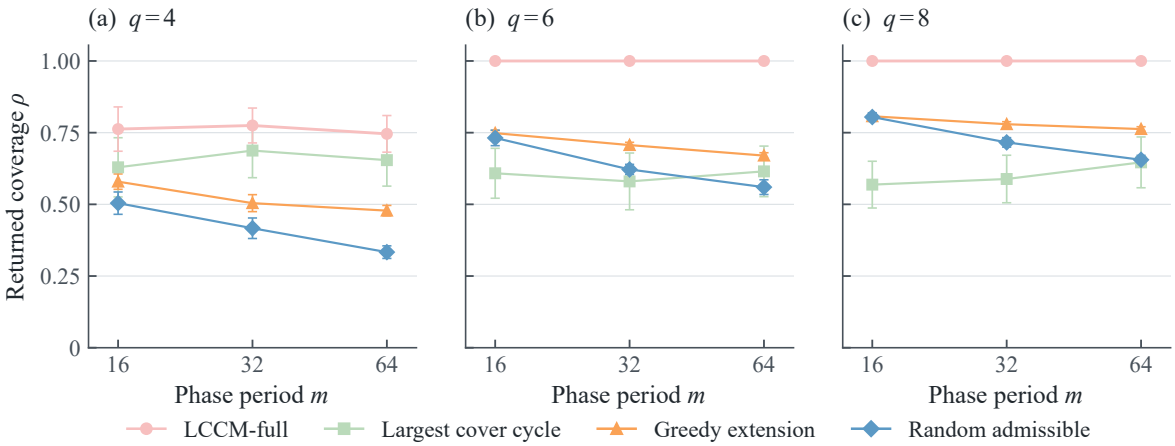



Figure 8.2: Coverage across channel counts. Each panel fixes $q$. Points show means and 95% confidence intervals over 20 unit weight seeds. Greedy extension at $q = 4$ and $m = 64$ has 19 valid cycles.

system should use multi-policy rotation or randomized exploration.

From a graph-theoretic perspective, it remains natural to determine the largest possible asymptotic perimeter gap among strongly connected vertex-transitive digraphs, and in particular whether the constant $1/6$ in our construction can be improved. It would also be interesting to identify additional structural conditions, such as degree or expansion assumptions, that force a spanning or near-spanning directed cycle. Bucić et al. [5] ask whether any two longest directed cycles in a connected vertex-transitive digraph must intersect. In our setting, this asks whether distinct maximum-coverage recurrent policies must share some controller states. They also ask whether there exists an absolute constant $C > 0$ such that, in every vertex-transitive digraph, the length of a longest directed path is at most $C$ times the circumference. This would clarify whether extensive transient exploration can coexist with substantially more limited recurrent coverage.

Several algorithmic and scalability limitations remain open. The exact mixed-integer formulation solves small instances to optimality but times out as the phase period $m$ grows. Meanwhile, LCCM serves as an efficient heuristic without a proven approximation ratio. Furthermore, our evaluation focuses on memory depth $k = 2$. Scaling to larger memory depths $k \geq 3$ creates exponentially larger state spaces with more complex first-return permutations. Developing approximation algorithms with theoretical performance bounds, stronger cutting planes for large periods, and multi-phase exchange neighborhoods offers clear directions for future algorithmic work.

**10 Conclusion** This paper introduces context coverage to quantify state space exploration under deterministic recurrent policies. Theoretically, we construct an infinite family of strongly connected vertex transitive digraphs with a coverage gap of at least $1/6$, yielding an asymptotic perimeter gap of at least $n/6$ and demonstrating that reachability and local symme-

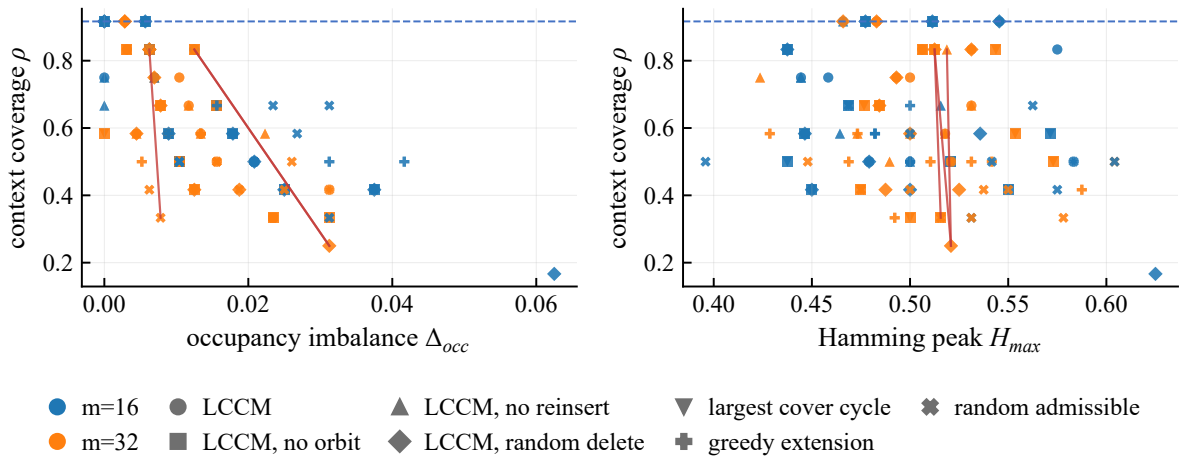



Figure 8.3: Sequence metrics versus context coverage in the $q = 4$ subset. Each marker is one validated run (shape: constructor; blue/orange: $m = 16/32$). Red segments link highlighted pairs satisfying (8.1). In both panels these pairs are nearly vertical: runs with almost identical occupancy imbalance $\Delta_{\mathrm{occ}}$ (left) or Hamming peak $H_{\max}$ (right) can differ in context coverage $\rho$ by up to $7/12 \approx 0.583$. The two sequence summaries therefore leave large coverage differences invisible.

Table 8.2: Quality, cost, and certification ($q = 4$, unit weights). $L$ and $B$ are the final MILP incumbent and upper bound; total runtime is not incumbent-discovery time. MILP uses generic runs for $m \leq 32$ and canonical E1 at $m = 64$. LCCM summarizes 20 seeds; intervals are 95% CIs.

| | | MILP: one recorded run | | | | LCCM-full: 20 seeds | |
|---|---|---|---|---|---|---|---|
| $m$ | $\lvert V\rvert$ | $L/\lvert V\rvert$ | $B/\lvert V\rvert$ | Total time (s) | Status | Mean $\rho$ (95% CI) | Mean time (s) |
| 4 | 48 | 0.9167 | 0.9167 | 16.1 | optimal | $0.721 \pm 0.072$ | 0.004 |
| 8 | 96 | 0.9167 | 1.0000 | 600.1 | timeout | $0.854 \pm 0.050$ | 0.010 |
| 12 | 144 | 0.9167 | 1.0000 | 600.1 | timeout | $0.829 \pm 0.057$ | 0.022 |
| 16 | 192 | 0.9167 | 1.0000 | 600.1 | timeout | $0.762 \pm 0.077$ | 0.032 |
| 24 | 288 | 0.9167 | 1.0000 | 600.2 | timeout | $0.762 \pm 0.081$ | 0.076 |
| 32 | 384 | 0.9167 | 1.0000 | 600.2 | timeout | $0.775 \pm 0.061$ | 0.121 |
| 64 | 768 | 0.9167 | 1.0000 | 600.5 | timeout | $0.746 \pm 0.064$ | 0.508 |

try do not ensure broad recurrent exploration. For periodic constraints, cyclic layering and first return orbit decompositions separate local capacity bottlenecks from global obstructions, underpinning both exact mixed integer formulations and the scalable LCCM heuristic. Computational evaluations confirm that conventional occupancy and Hamming summaries can conceal coverage discrepancies as large as 0.583, whereas LCCM consistently recovers full coverage across all tested six and eight channel instances. Beyond single cycle execution, these structural insights motivate multi policy operation and selective constraint relaxation to recover omitted contexts. Extending this state space framework to broader asymmetric network topologies represents a natural next step. Together, these findings establish context coverage as both a structural diagnostic and an optimization objective, demonstrating that comprehensive recurrence requires optimizing over the complete controller state graph rather than the emitted sequence alone.

**Artifact and Reproducibility** An anonymized artifact is available at https://anonymous.4open.science/r/HCPH-783B. It contains the instance generators, algorithm implementations (MILP, LCCM, and the baseline), evaluation scripts, test suites, data, and environment configurations.

**Declaration of Generative AI Usage** During paper preparation, the authors used LLMs to assist with writing, coding, and heuristic suggestions for intermediate proof steps. All AI-assisted content was reviewed, tested, and verified by the authors. The authors assume full responsibility for all content.

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## A Appendix

**A.1 Proof of the Coverage Bound** This section gives the complete proof of Theorem 4.1. We use the notation $r$ and $M = 2^r$ from Section 4; the phase period of the computational model remains denoted by $m$.

*Proof of Theorem 4.1.* Let $M = 2^r$, let $R = \mathbb{Z}[\omega]$, where $\omega^2 + \omega + 1 = 0$, and let $A = R/MR$. Then $|A| = M^2$. Let $\lambda = 1 + \omega$. We have $\lambda^2 = \omega$, $\lambda^3 = -1$, and $\lambda^6 = 1$.

Let $\Gamma = A \rtimes \mathbb{Z}_6$, where $\mathbb{Z}_6$ acts on $A$ by multiplication by $\lambda$. Thus $(x, i)(y, j) = (x + \lambda^i y, i + j)$, with the second coordinate taken modulo 6. Let $a = (0, 2)$ and $b = (1, 3)$, and let $D$ be the directed Cayley graph of $\Gamma$ with arcs $g \to ga$ and $g \to gb$.

We first prove that $D$ is strongly connected. Direct calculation gives $a^3 = b^2 = (0, 0)$, $aba^{-1} = (\omega, 3)$, $b(aba^{-1}) = (1 - \omega, 0)$ and $a(b(aba^{-1}))a^{-1} = (\omega(1 - \omega), 0)$. The elements $1 - \omega$ and $\omega(1 - \omega)$ generate the ideal $(1 - \omega)R$ as an additive group. Moreover, $(1 - \omega)(1 - \omega^2) = 3$. Since $M$ is a power of 2, 3 is invertible modulo $M$, and multiplication by $1 - \omega$ is an automorphism of $A$. Hence $a$ and $b$ generate every translation in $A$. Their images in $\mathbb{Z}_6$ are 2 and 3, which generate $\mathbb{Z}_6$. Therefore $\langle a, b\rangle = \Gamma$. Since $a^{-1} = a^2$ and $b^{-1} = b$, the digraph $D$ is strongly connected. Left multiplication by $\Gamma$ is transitive on $V(D)$. Since $a$ and $b$ are distinct nonidentity elements, $D$ has no loops or parallel arcs.

The $a$-arcs partition $V(D)$ into directed triangles $\{(x, e), (x, e + 2), (x, e + 4)\}$, where $x \in A$ and $e \in \{0, 1\}$. The $b$-arcs are the two orientations of a perfect matching. Contract every $a$-triangle. Denote by $E_x$ the triangle with even second coordinates and by $O_x$ the triangle with odd second coordinates. The resulting bipartite graph $H$ has edges $E_xO_{x+1}$, $E_xO_{x+\omega}$, and $E_xO_{x+\omega^2}$ for every $x \in A$. These three neighbors are distinct, so $H$ is simple.

Let $L = (1 - \omega)R$. Consider the plane graph with black vertex set $L$ and white vertex set $1 + L$, in which each $z \in L$ is joined to $z + 1$, $z + \omega$, and $z + \omega^2$. Together with its hexagonal faces, this graph defines a cell decomposition of $\mathbb{C}$.

Reduction modulo $MR$ maps $L$ onto $A$, with kernel $ML$, because $1 - \omega$ is invertible modulo $M$. Every nonzero vector of $ML$ has length at least $M|1 - \omega| = M\sqrt{3} > 2$, while each closed hexagonal face has diameter 2. Thus no two distinct points of a closed face differ by an element of $ML$, and the quotient map is injective on every closed face. Translations by vectors in $ML$ preserve this cell decomposition. Hence its quotient on $\mathbb{C}/ML$ has $H$ as its 1-skeleton, and every face is bounded by a simple 6-cycle. Since $H$ is cellularly embedded in the torus, which has genus 1, Euler's formula gives $|V(H)| - |E(H)| + |F(H)| = 0$. As $|V(H)| = 2M^2$ and $|E(H)| = 3M^2$, it follows that $|F(H)| = M^2$.

Let $C$ be a directed cycle in $D$. Because cycles formed exclusively of $a$-arcs or $b$-arcs have lengths 3 and 2, respectively, we may assume that $C$ contains both types of arcs. Between two consecutive $b$-arcs, $C$ traverses either one or two $a$-arcs, since omitting $a$-arcs produces a 2-cycle along a matching edge, while traversing three $a$-arcs prematurely returns to the initial vertex.

Each visit of $C$ to an $a$-triangle therefore contains at least two of its three vertices. If $C$ visited the same $a$-triangle twice, the two visits would share a vertex, a contradiction. Hence the $a$-triangles visited by $C$ are distinct. After these triangles are contracted, the $b$-arcs of $C$ form a closed walk through distinct vertices of $H$. This walk cannot have length 2, since that would force $C$ to use the same matching edge twice and repeat one of its end vertices. It is therefore a simple cycle; denote it by $Q$.

Let $\ell = |Q|$, and let $q$ be the number of visits at which $C$ uses two $a$-arcs. The cycle $C$ contains $\ell$ matching arcs, $\ell - q$ visits using one $a$-arc, and $q$ visits using two $a$-arcs. Hence

$$|C| = \ell + (\ell - q) + 2q = 2\ell + q.$$

Suppose that $C$ enters an $a$-triangle at $(x, i)$. In the plane embedding, the entering matching edge has direction $-\lambda^i = \lambda^{i+3}$. If $C$ uses one $a$-arc, it leaves along a matching edge of direction $\lambda^{i+2}$, so the signed turn of $Q$ is $-\pi/3$. If $C$ uses two $a$-arcs, it leaves in direction $\lambda^{i+4}$, so the signed turn is $\pi/3$. Thus $q$ vertices of $Q$ have signed turn $\pi/3$, and the other $\ell - q$ vertices have signed turn $-\pi/3$.

Suppose that $Q$ is non-contractible (if, as a map $Q \cong S^1 \to \mathbb{C}/ML$, it is not homotopic to a constant map). Let $\gamma$ be a smoothing of $Q$, and define its tangent map by

$$\tau_\gamma(t) = \frac{\gamma'(t)}{|\gamma'(t)|} \in S^1.$$

Since $\gamma$ is isotopic to a closed geodesic $\gamma_0$, $\deg(\tau_\gamma) = \deg(\tau_{\gamma_0}) = 0$. Hence, we have

$$\frac{(2q - \ell)\pi}{3} = 2\pi \deg(\tau_\gamma) = 0,$$

and therefore $q = \ell/2$. Since $\ell \le |V(H)| = 2M^2$,

$$|C| = 2\ell + q = \frac{5\ell}{2} \le 5M^2.$$

It remains to consider a contractible cycle $Q$. Its lift $\widetilde{Q}$ under $\mathbb{C} \to \mathbb{C}/ML$ is a simple closed polygon in the plane. Hence the sum of the signed exterior angles of $\widetilde{Q}$ is $2\pi$ or $-2\pi$, according to its orientation. Therefore

$$q\frac{\pi}{3} - (\ell - q)\frac{\pi}{3} = \pm 2\pi,$$

and hence $|2q - \ell| = 6$.

We first prove that $Q$ is not Hamiltonian. Suppose that $Q$ contains all $2M^2$ vertices of $H$. As a contractible simple closed curve on the torus, $Q$ bounds a disk $\Delta$. Every vertex of $H$ lies on $\partial\Delta = Q$, so the interior of $\Delta$ contains no vertex of $H$. The hexagonal faces contained in $\Delta$ form a cell decomposition of $\Delta$. Let $e_{\text{int}}$ and $f$ be the numbers of interior edges and faces, respectively. If $Q$ is Hamiltonian, then $|V(Q)| = |E(Q)| = 2M^2$. Euler's formula for the disk yields $2M^2 - (2M^2 + e_{\text{int}}) + f = 1$, so $e_{\text{int}} = f - 1$. Counting the sides of the hexagonal faces yields $6f = 2e_{\text{int}} + 2M^2$. Hence $6f = 2(f-1) + 2M^2$, and therefore $f = (M^2 - 1)/2$. Since $M$ is even, $(M^2 - 1)/2 \notin \mathbb{Z}$, a contradiction. Thus $Q$ is not Hamiltonian.

The graph $H$ is bipartite, with $M^2$ vertices in each part. Therefore a non-Hamiltonian cycle in $H$ has length at most $2M^2 - 2$, so $\ell \leq 2M^2 - 2$. Also, $|2q - \ell| = 6$ gives $q \leq (\ell + 6)/2$. Thus,

$$|C| = 2\ell + q \leq \frac{5\ell + 6}{2} \leq 5M^2 - 2.$$

Thus every directed cycle in $D$ has length at most $5M^2$. Since $|V(D)| = 6M^2$, the perimeter gap of $D$ is at least

$$6M^2 - 5M^2 = M^2 = \frac{|V(D)|}{6}. \qquad \square$$

**A.2 Circumference Bound on Small Instances** Table A.1 evaluates the $5M^2$ bound on the first three Cayley instances. The MILP certifies optimality for $r \leq 2$. For $r = 3$, an independently constructed cycle achieves length $320 = 5M^2$, confirming bound tightness despite solver timeout.

Table A.1: Circumference on small Cayley instances.

| $r$ | $M$ | $\lvert V\rvert$ | $L$ | $5M^2$ | $L/(5M^2)$ | Status |
|---|---|---|---|---|---|---|
| 1 | 2 | 24 | 20 | 20 | 1.000 | optimal |
| 2 | 4 | 96 | 80 | 80 | 1.000 | optimal |
| 3 | 8 | 384 | 320 | 320 | 1.000 | witness cycle |

**A.3 Experimental Protocol and Statistical Methods** All evaluations are conducted on the cyclically layered phase history graph described in Section 3.

Table A.2: LCCM variants used in the primary ablation.

| Variant | Merge objective | Repair | Reinsertion |
|---|---|---|---|
| Full | orbit count | guided | enabled |
| No orbit | largest component | disabled | enabled |
| No reinsert | orbit count | guided | disabled |
| Random delete | orbit count | random orbit | enabled |

The primary testbed systematically evaluates seven constructors, comprising full LCCM, three ablation variants, the largest initial cycle cover, greedy extension, and random admissible extension, across channel counts $q \in \{4, 6, 8\}$, phase periods $m \in \{16, 32, 64\}$, and 20 random seeds. For each seed, generated trajectories are evaluated under both unit weight and hashed weight regimes. Coverage statistics summarize certified feasible cycles, while reported runtimes reflect total execution across all attempts.

Statistical comparisons are reported as sample means with 95% Student $t$ confidence intervals given by $\bar{x} \pm t_{0.975,n-1} s/\sqrt{n}$. Paired intervals quantify differences between constructors across matched seeds, evaluated using $n = 20$ seeds in all settings except for greedy extension at $q = 4$ and $m = 64$, where $n = 19$ due to construction failure.

**A.4 LCCM implementation and variants** The LCCM method starts by building a balanced matching system. Full canonical layers use seeded cyclic permutations within middle channel groups. Other layers find augmenting paths on sorted state lists and apply seeded shuffling. The merge stage connects cycles through valid two edge exchanges at a single phase boundary.

When merging leaves multiple first return orbits, the repair step removes the smallest orbit. Ties are broken by state order. The algorithm then rematches the remaining states. It accepts the new cover if the largest cycle grows or if the orbit count drops. Next, reinsertion attempts to increase occupancy to $h + 1$. This candidate is accepted only if the largest cycle strictly expands. At the end, the algorithm returns the largest cycle found in the final cover.

Table A.2 summarizes the algorithm variants. Full LCCM guides both the merge step and the repair step using orbit counts. The no orbit variant merges components by size and disables repair. The no reinsert variant keeps repair but skips occupancy growth. The random delete variant removes a random orbit during repair instead of the smallest one.

All primary heuristic runs declare a 60-second bud-

Table A.3: Cut-free canonical phase-history runs. Here $|V| = 12m$ and $|E| = 24m$.

| $m$ | $\|V\|$ | $L$ | $B$ | gap (%) | status |
|---|---|---|---|---|---|
| 2 | 24 | 22 | 22 | 0.0 | optimal |
| 3 | 36 | 33 | 33 | 0.0 | optimal |
| 4 | 48 | 44 | 44 | 0.0 | optimal |
| 5 | 60 | 55 | 60 | 9.1 | timeout |
| 6 | 72 | 66 | 72 | 9.1 | timeout |
| 8 | 96 | 88 | 96 | 9.1 | timeout |
| 10 | 120 | 110 | 120 | 9.1 | timeout |
| 12 | 144 | 132 | 144 | 9.1 | timeout |
| 16 | 192 | 176 | 192 | 9.1 | timeout |
| 24 | 288 | 264 | 288 | 9.1 | timeout |
| 32 | 384 | 352 | 384 | 9.1 | timeout |
| 48 | 576 | 528 | 576 | 9.1 | timeout |
| 64 | 768 | 704 | 768 | 9.1 | timeout |

get, 4000 candidates per search round, and 20 construction attempts. The initial merge has a 250-round limit; merges after repair and reinsertion each have a 125-round limit. The time check occurs during merge searches and at stage boundaries, so it does not interrupt a matching construction. Path-extension baselines use at most 32 starts and at most $|V|$ steps per start. A common validator checks state uniqueness, every transition, and the closing arc.

**A.5 MILP** The flow formulation fixes a root and optimizes over cycles that pass through it. For the unrestricted canonical graph with unweighted cardinality, verified vertex transitivity allows a single canonical root. Otherwise the solver tries each root separately. A result is labeled optimal when a valid incumbent exists and the global bound is closed across all root subproblems. A timeout incumbent is a feasible lower bound. All exact runs use one solver thread and a 600 s limit unless stated otherwise.

Table A.3 lists the thirteen cut-free canonical records. Every row has an incumbent of length $11m$. The solver closes the optimization gap for $m = 2, 3, 4$, whereas for the remaining periods its internal upper bound remains at the trivial value $12m$. These open solver gaps do not represent unresolved mathematical gaps: by the Li–Methuku upper bound [16], the circumference of this phase-history graph is at most $11m$. Hence every recorded $11m$ incumbent is globally optimal. The independent SciPy/HiGHS replay in Figure 8.1 produces an 88-state witness for $m = 8$, thereby attaining this theoretical upper bound. The Cayley bound of Theorem 4.1 concerns a different graph family and is not used in this certification.

The generic and layer-aware formulations both use one thread and a 600 s budget in Table A.4. Layer balance cuts the $m = 4$ solve time and node count. The larger paired runs all time out.

**A.6 LCCM Ablation** Table A.5 reports the unsaturated $q = 4$ settings. The paired mean differences between LCCM-full and no-orbit at $m = 16, 32, 64$ are 0.1333, 0.0875, and 0.0917. The unadjusted 95% intervals are $[0.0435, 0.2232]$, $[0.0121, 0.1629]$, and $[0.0311, 0.1522]$. The no-orbit variant also disables repair, so these differences reflect the combined effect of orbit-guided merging and repair rather than either one alone. The reinsertion and deletion-rule comparisons are less stable because several paired intervals include zero. All four variants reach full coverage at every tested $q = 6, 8$ setting, so no component-level comparison is possible there.

**A.7 Sensitivity to Initial Cover** A separate 480-record cohort compares the full seeded initializer with a degree-ordered greedy initializer under the same later stages and budgets. At $q = 4$, the mean coverage differences, full minus greedy, are 0.0167 for $m = 16$ and 0.0125 for $m = 32$. The paired 95% intervals are $[-0.0640, 0.0973]$ and $[-0.0754, 0.1004]$. Both include zero. Both initializers reach full coverage at the tested $q = 6, 8$ settings. Neither initializer shows a reliable advantage.

**A.8 Empirical Search Dynamics and Traces** The instrumented cohort contains 840 records at $q = 4$ and $m \in \{16, 32, 64\}$. It records a timestamp and validated witness for every strict improvement. All terminal digests and statuses match the primary records. In 42 records the best observed cycle is longer than the terminal returned cycle. This happens because repair sometimes accepts fewer states to reduce the orbit count. We therefore keep terminal returned coverage and best-observed coverage as separate quantities. Values before the first valid discovery are left blank rather than set to zero.

**A.9 Representative Case on Metric Blind Spots** The $q = 4$, $m \in \{16, 32\}$ subset contains 9660 eligible unordered pairs within common $(q, m,$ weight mode) groups. Of these, 2092 satisfy (8.1). Table A.6 shows three representative examples. These counts describe run-pair comparisons, not independent statistical tests or distinct trajectories.

Table A.4: Paired generic and layer-aware MILP records. Timeout rows report validated incumbents, solver upper bounds, and open gaps.

| $m$ | formulation | incumbent $L$ | bound $B$ | gap (%) | time (s) | nodes | status |
|---|---|---|---|---|---|---|---|
| 4 | generic | 44 | 44 | 0.0 | 16.1 | 45948 | optimal |
| 4 | layer-aware | 44 | 44 | 0.0 | 3.2 | 20791 | optimal |
| 8 | generic | 88 | 96 | 9.1 | 600.1 | 1015564 | timeout |
| 8 | layer-aware | 88 | 96 | 9.1 | 600.1 | 4721175 | timeout |
| 12 | generic | 132 | 144 | 9.1 | 600.1 | 598733 | timeout |
| 12 | layer-aware | 132 | 144 | 9.1 | 600.1 | 743692 | timeout |
| 16 | generic | 176 | 192 | 9.1 | 600.1 | 355449 | timeout |
| 16 | layer-aware | 176 | 192 | 9.1 | 600.2 | 417560 | timeout |
| 24 | generic | 264 | 288 | 9.1 | 600.2 | 236140 | timeout |
| 24 | layer-aware | 264 | 288 | 9.1 | 600.2 | 1617242 | timeout |
| 32 | generic | 352 | 384 | 9.1 | 600.2 | 128408 | timeout |
| 32 | layer-aware | 352 | 384 | 9.1 | 600.3 | 1207905 | timeout |

Table A.5: Coverage ablation for $q = 4$: mean $\pm$ 95% confidence interval over 20 unit-weight seeds.

| $m$ | LCCM-full | no-orbit | no-reinsert | random-delete |
|---|---|---|---|---|
| 16 | $0.762 \pm 0.077$ | $0.629 \pm 0.103$ | $0.729 \pm 0.075$ | $0.650 \pm 0.116$ |
| 32 | $0.775 \pm 0.061$ | $0.688 \pm 0.095$ | $0.729 \pm 0.071$ | $0.667 \pm 0.127$ |
| 64 | $0.746 \pm 0.064$ | $0.654 \pm 0.091$ | $0.692 \pm 0.081$ | $0.625 \pm 0.108$ |

Table A.6: Representative run pairs ($m = 32$) satisfying (8.1). Each pair is shown as two run records.

| pair | mode | policy (seed) | $\Delta_{\mathrm{occ}}$ | $H_{\max}$ | $\rho$ |
|---|---|---|---|---|---|
| 1 | hashed | LCCM, no reinsert ($s = 5$) | 0.0125 | 0.5188 | 0.8333 |
| 1 | hashed | LCCM, random delete ($s = 1$) | 0.0312 | 0.5208 | 0.2500 |
| 2 | hashed | LCCM, random delete ($s = 1$) | 0.0312 | 0.5208 | 0.2500 |
| 2 | hashed | largest cover cycle ($s = 5$) | 0.0125 | 0.5125 | 0.8333 |
| 3 | hashed | LCCM ($s = 8$) | 0.0063 | 0.5125 | 0.8333 |
| 3 | hashed | random admissible ($s = 5$) | 0.0078 | 0.5156 | 0.3333 |

Table A.7: Solver records used for the recurrence plateau comparison. Total time is solver runtime.

| $m$ | $\lvert V\rvert$ | $L$ | bound $B$ | gap (%) | time (s) | status |
|---|---|---|---|---|---|---|
| 4 | 48 | 44 | 44 | 0.0 | 16.8 | optimal |
| 8 | 96 | 88 | 96 | 9.1 | 327.4 | feasible |
| 16 | 192 | 176 | 192 | 9.1 | 600.1 | timeout |

**A.10 Recurrence Plateaus and Uniform Randomized Exploration** We take validated cycles from separate cut-free solver runs at $m = 4, 8, 16$ and simulate 30 uniform-action random walks with seeds $0, \ldots, 29$ per instance. Each walk starts from a uniformly sampled state and runs for $20|V|$ controller updates. We define the first-reach time as the first one-based sample whose cumulative coverage reaches at least $11/12$. The mean first-reach times are 112.8, 213.9, and 448.7 steps for $m = 4, 8, 16$. Every sampled walk eventually exceeds $11/12$. The MILP solver itself certifies optimality for $m = 4$, but the $11m$ cycles for $m = 8$ and $m = 16$ are also globally optimal by the Li–Methuku upper bound [16]. Thus the deterministic reference level $11/12$ is the exact maximum recurrent coverage for all three instances.

**A.11 Robustness under Structural Perturbations** The 200 perturbation cells fall into two families. The first is 120 random-edge deletion cells at $q = 4$ with $m \in \{16, 32\}$. The second is 80 phase-blacklist cells at $q = 16$ with $m \in \{17, 31\}$. Among all cells, 125 have positive matching deficiency, 122 have a largest SCC ratio below one, and eight returned covers contain more than one first-return orbit. These categories overlap. A valid cycle is returned in 75 cells. The other 125 runs end with `CoverError` within the construction budget. A construction failure does not prove that no feasible cycle exists. Similarly, a fragmented returned cover does not prove that every compatible matching system is fragmented.